HACKINSIGHT 2026

# FloodLens Pakistan

District heavy-rain warnings 1–3 days ahead, with a weekly food-price watch

**Team Spark H** | Climate & disaster risk · food security

huggingface.co/spaces/RoshaanT1/floodlens-pakistan



160 districts · scored every hour

# Problem Statement

THE PROBLEM

Monsoon rain in Pakistan turns deadly within hours. District planners must decide where to send rescue teams, pumps and relief before the rain arrives, but weather models give grid cells, not a per-district answer to “is this unusual for us?”

**1,037**

deaths in the 2025 monsoon (NDMA final sitrep No. 87)

1

## Who is affected?

District and PDMA disaster planners, and the people in Pakistan's 160 districts they serve

2

## Why does it matter?

Planners need 24–72 hours of warning per district to pre-position teams and stock

3

## Our goal

A calibrated chance of heavy rain for every district, 1–3 days ahead, better than climatology

# Dataset Declaration

| | |
|---|---|
| **Dataset name** | FloodLens district-day warehouse (5 public sources joined on 160 districts) |
| **Source / link** | Open-Meteo Previous Runs (API) · NASA GPM IMERG V07 · OCHA boundaries (HDX) · PBS weekly SPI · NDMA sitreps |
| **Size** | 476,640 forecast rows × 25 features · 689,280 satellite rain rows · 146,523 prices · 157 NDMA documents |
| **Time period** | Rain 2015–2026 · forecasts Jul 2024–Oct 2026 · prices Jan 2023–Oct 2026 |
| **Format & license** | Raw files + SHA-256 manifest → Parquet / DuckDB · CC BY 4.0, NASA open data, public govt. statistics |
| **Key features / target** | Forecast rain, humidity, pressure, cloud, wind (1–3 days ahead), lagged observed rain. Target: rain above the district's own 95th percentile |

**476K**
Rows / records (model table)

**25**
Columns / features

**1.3 GB**
Raw data, 4,856 files

# Pipeline Used

1

## Data Collection

APIs, NASA cloud, PBS and NDMA files; every file hashed into a manifest

2

## Data Cleaning

Quality agent: fix, fill or flag each issue, with the reason logged

3

## Analysis / EDA

District thresholds, monsoon replay, error by province and terrain

4

## Modeling

Gradient boosting, calibrated; beats climatology on the locked test set

5

## Dashboard / App

Hourly Prefect flow publishes to a live Hugging Face Space

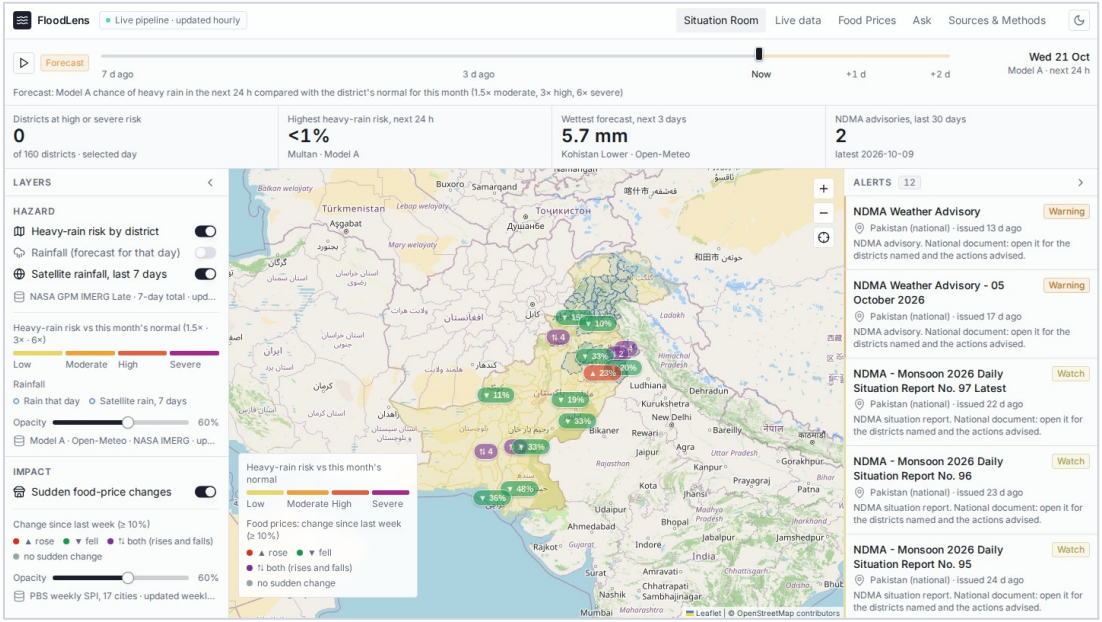
**Raw forecasts and satellite rain in → each district's chance of unusually heavy rain 1–3 days ahead, refreshed every hour**

# Pipeline Details & Tools

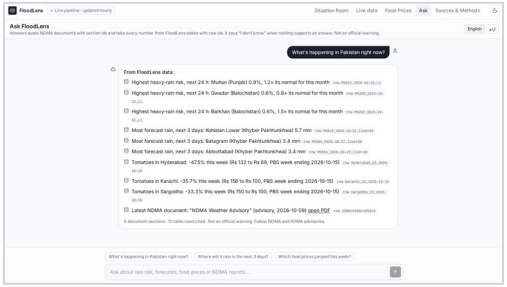
| Stage | Tools / Libraries | What we did |
|---|---|---|
| **Collection** | Python, requests, earthaccess (OPeNDAP), pdfplumber | Fetch only what is new; respect API budgets; never overwrite a raw file |
| **Preprocessing** | DuckDB, pandas, Shapely, pandera | Grid rain → district values; quality rules fix / fill / flag and log every decision |
| **Analysis / EDA** | pandas, bootstrap CIs | 95th-percentile thresholds per district; Aug 2025 replay; skill by province and terrain |
| **Modeling** | scikit-learn GBM + Platt calibration, FAISS + e5 | Leakage-tested features; fixed splits; test set run once. Insight Agent cites NDMA text |
| **Dashboard / App** | Prefect 3, React + Leaflet, FastAPI, HF Spaces | Hourly flow rebuilds and publishes; map, food prices, live data and Ask (English / Urdu) |

# Dashboard / App Views

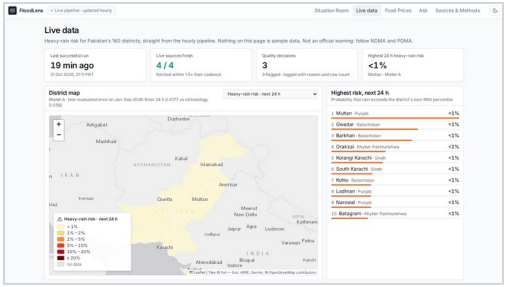
## Page 1 – Situation Room: risk vs each district's normal




## Page 2 – Ask (Insight Agent)



## Page 3 – Live data



**Live demo:** huggingface.co/spaces/RoshaanT1/floodlens-pakistan • **Repo:** github.com/RoshaanT1/floodlens-pakistan

# Conclusion

1

## Beats the baseline on unseen 2026 data

Brier 0.0177 vs climatology 0.0192 (about 7% better); the 95% interval of the gain excludes zero

2

## Better everywhere we checked

Ahead of climatology in every province and terrain group and in monsoon months; weakest in Azad Kashmir

3

## Right when it mattered

Replaying Aug 2025: on 14 Aug, 8 of the 10 districts we ranked highest really had heavy rain

### IMPACT

160 district offices and their PDMAs get a free, hourly, per-district signal with the reasons and NDMA sources one click away.

### FUTURE WORK

- Sub-daily rain to catch cloudbursts like Buner
- Flood and damage targets, not only rain
- A price model that beats “no change”

# Team & Credits

RT

## Roshaan Tahir

**Forecast data, models & dashboard**

- Open-Meteo forecast pipeline
- Model A training and calibration
- Insight Agent and live dashboard

HK

## Hammad Khan

**Satellite rain & ground truth**

- NASA IMERG satellite rain fetcher
- District rain, 95th-pct thresholds, labels
- Data quality checks + 26 tests

MH

## Muhammad Huzaifa

**Deployment & delivery**

- Hugging Face Space (Docker server)
- Build-and-assemble deploy script
- Deploy checklist and AI-use records

**Contact:** github.com/RoshaanT1/floodlens-pakistan · Data: Open-Meteo, NASA, OCHA, PBS, NDMA

# Thank you!